

Humidification

Pre-Lab Plan

Unit Operations Lab 6

11/11/25

Group 3, Tuesday, AM section

Hussein Rezk

Adil Feratovic

Aaron Casas

Anna Cai

Experimental Objective (5 pts)

The goal of this lab is to determine both heat and mass transfer by adjusting pressure and keeping the flowrate of water constant. This is done via a wet wall humidification column, adjusting inlet flowrate at room temperature, moreover, the same will be done with steam in a higher temperature state. By recording the air and water temperatures as well as humidity transfer coefficients can be calculated and compared to literature.

Experimental

Apparatus (5 pts)

The wetted-wall humidification unit consists of a 2-inch diameter Plexiglas column that is 4 feet long and connected to both an air flow system and a water feed system. Air is supplied from the house line, and its flow rate is controlled by a rotameter before entering the bottom of the column. Water is circulated from a holding tank using a small centrifugal pump, with its flow rate also measured by a rotameter. The water enters the top of the column, where a weir distributes it evenly along the inner wall to form a thin film. In this configuration, air and water flow countercurrently within the column.

The temperature of the incoming air is measured with a thermocouple, while the temperature and humidity of the exiting air are determined using either a wet-bulb thermometer or an electronic hygrometer located at the top of the column. When the system is operated without water, it provides the baseline humidity of the incoming air.

The water temperature is recorded at both the pump outlet and the holding tank using thermocouples. Additionally, three thermocouples are mounted along the column wall to monitor the temperature profile of the descending water film. The system is also equipped with a heat exchanger that allows the incoming air to be preheated, enabling experiments to be conducted at various air temperatures and humidification rates.

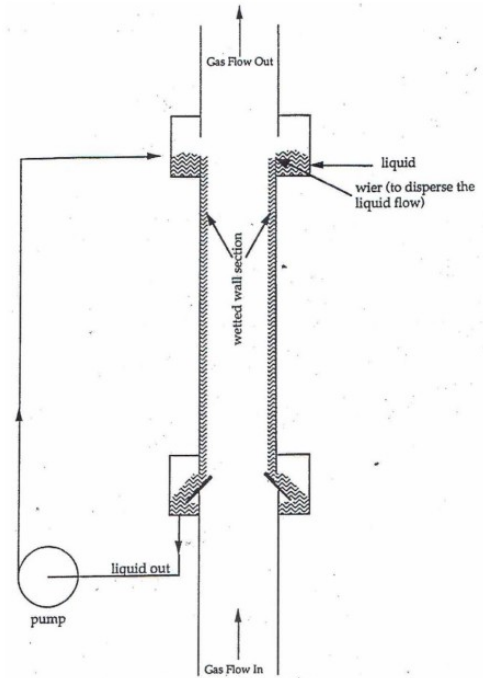


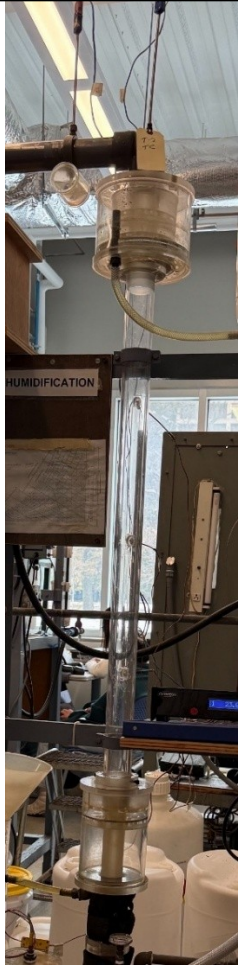
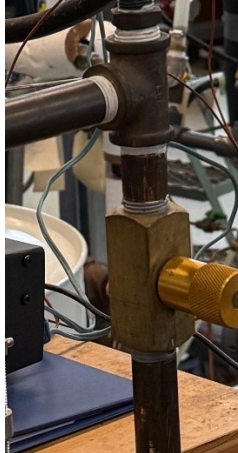
Figure 1: Schematic of a generalized wetted wall column

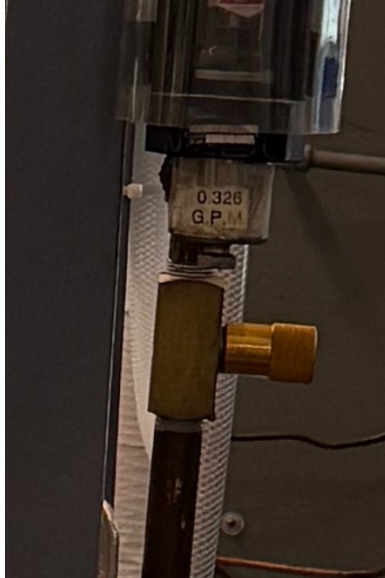


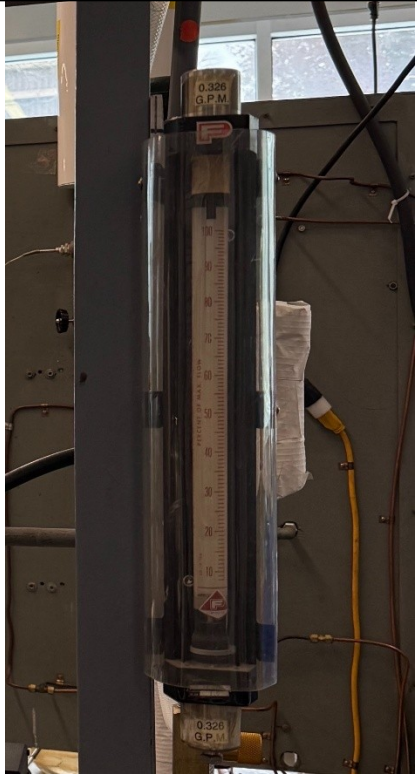
Figure 2: Main apparatus.

Table 1: List of Main Apparatus Equipment and Description

Equipment	Picture	Description
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Humidification Column		Water is pumped into this plexiglass column to wet the inner surface, and air is sent up the center to humidify the water.
Air Flowrate Valve		Controls the flowrate of air entering the humidification column.

Air flowrate gauge	 A vertical air flowrate gauge with a black top section and a clear cylindrical body showing a scale. It is mounted on a metal frame in a laboratory setting with a window in the background.	Gauge that measures the flowrate of air
Water Flowrate valve	 A close-up of a water flowrate valve. It features a clear cylindrical body with a yellow adjustment knob on the side. A label on the device reads "0.326 GPM". It is connected to a white braided hose.	Controls the flowrate of water in the column.

Water Flowrate Gauge		Shows the flowrate of water within the column.
Air pressure gauge and dial		Controls the air pressure delivered to the system.

Thermocouples and
Thermocouple Reader



Six thermocouples (3 on the column wall, one at bubble-point, one at dry-bulb, and one in the reservoir) and the reader that displays the measured temperatures.

Steam Heat Exchanger		Heat exchanger used to heat the air entering the column, the valve controls the amount of steam sent to the heat exchanger.
Water pump		Pump that moves the water out of the reservoir and into the column.
Pump switch		Turns the pump on and off.

Water reservoir		The tank used to store water being recycled through the column in the system.
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Materials and Supplies (5 pts)

Table 2: List of Materials and Supplies

Materials	Description
DI Water	Used to keep reservoir full and ensure column is uniform.
Steam	Steam generator for temperature increase
Beakers (1-4L)	Used for water preparation and transfer
Plastic Funnel	Used for safe transfer of water to system
Heat resistant gloves	Used when handling hot steam
Metal scissor jack	Support for the outlet and adjustable height of steam meter
Air	Fed from air house supply
2-Step Ladder	Used to elevate to outlet and read humidification
Anemometer/Hygrometer	Used to quantify % humidification in the outlet

Experimental Plan (10 pts)

Procedure:

- Verify the thermocouples
- Confirm rotameter readings
- With no water flow, measure the inlet air temperature
- Start the water flow and confirm continuous film and stable wall temperature profile
- Choose temperature then set target air rotameter to hit desired Re
- Once steady state is reached, record inlet air temperature, outlet gas temperature and RH, wall thermocouples, water tank temperature, air and water flow rates, and steam settings
- Repeat with different temperatures
- Take replicate data

Variables:

Independent: Air flowrate, inlet air temperature, and water flow rate

Dependent: Outlet gas humidity, kY, h, NTU, HTU, Sh, Re, and Sc

Statistical/Calibration Methods:

- Take at least 1 set of replicate data for each run
- Confidence intervals: Use 95% confidence on kY and h for each run
- Goodness of fit: Find R^2 for $Sh = aRe^b Sc^c$
- Plot h and kY as a function of Re
- Plot the Sherwood number divided by $Sc^{0.44}$ vs Re on a log plot

Correlation with Theory:

- Experimentally derived kY and h will be compared against theoretical predictions
- The Lewis relationship, $1 = \frac{h}{k_Y C'_S}$, will be tested
- The adiabatic saturation temperature and humidity ratio readings will be validated by comparing results gained via a psychrometric chart. This will ensure that humidity ratio and temperature values gathered are consistent with natural physical laws.

Project Safety Evaluation

Project Safety Evaluation (10 pts)

	Yes/No	
Potential electrical hazards?	Yes	If yes, identify conditions: The pump and the thermocouple operate on electricity. The experiment itself has water circulating, so naturally, this poses a hazard to the pump as well as to the thermocouples. If water is filled too quickly into the funnel, water can potentially spill onto the pump.
Potential mechanical hazards?	Yes	If yes, identify conditions: If the pump is running dry, as in no water is circulating while it's operating, it can cause mechanical failure, causing the pump to break down.
Potential pressure hazards?	Yes	If yes, identify conditions: The compressed air enters at roughly 120 psi, traveling to a pressure regulator posing a potential risk if the regulator were to fail.
Potential temperature hazards?	Yes	If yes, identify conditions: Steam is used throughout the experiment, which reaches a temperature of roughly 113 degrees Celsius.
Hazardous/toxic chemicals involved?	no	If yes, hazards involved:
Gloves required?	yes	If yes, specify type: nitrile gloves as well as heat-resistant gloves.
MSDS reviewed by all team members?	Yes	List MSDS sheets reviewed: DI water and steam (7732-18-5)
<i>Process Parameters</i>	<i>Max Expected</i>	<i>List location and possible hazards and precautions</i>
Temperature (°C)	113	The steam being supplied from the heat exchanger is at very high temperatures; therefore, heat-resistant gloves are required when operating the valves.
Pressure (psia)	134.7	The air hose supplies air entering at 120 psi, which is regulated with a pressure regulator. If the pressure regulator fails the full 120 psi pressure would enter the system.
<i>Safety Controls</i>	<i>yes/no</i>	<i>If yes, Provide description</i>
<i>Pressure relief</i>	<i>yes</i>	<i>Use a pressure relief valve and regulator to reduce the 120 psi supply gradually.</i>
<i>Steam shutoff valve</i>		<i>Use the steam shutoff valve when adjusting the heat exchanger temperature or shutting it off completely.</i>
<i>Shut of switch</i>		<i>A switch is present to shut of the entire process</i>

I have read relevant background material for the Unit Operations Laboratory entitled:

“_____” and understand the hazards associated with conducting this experiment. I have planned out my experimental work in accordance to standards and acceptable safety practices and will conduct all of my experimental work in a careful and safe manner. I will also be aware of my surroundings, my group members, and other lab students, and will look out for their safety as well.

<u>Hussein Rezk</u>	signature of team member	<u>11/11/25</u> date
<u>Aaron Casas</u>	signature of team member	<u>11/11/25</u> date
<u>Adil Feratovic</u>	signature of team member	<u>11/11/25</u> date
<u>Anna Cai</u>	signature of team member	<u>11/12/25</u> date